



TrioDocs

Version: 0.4.0

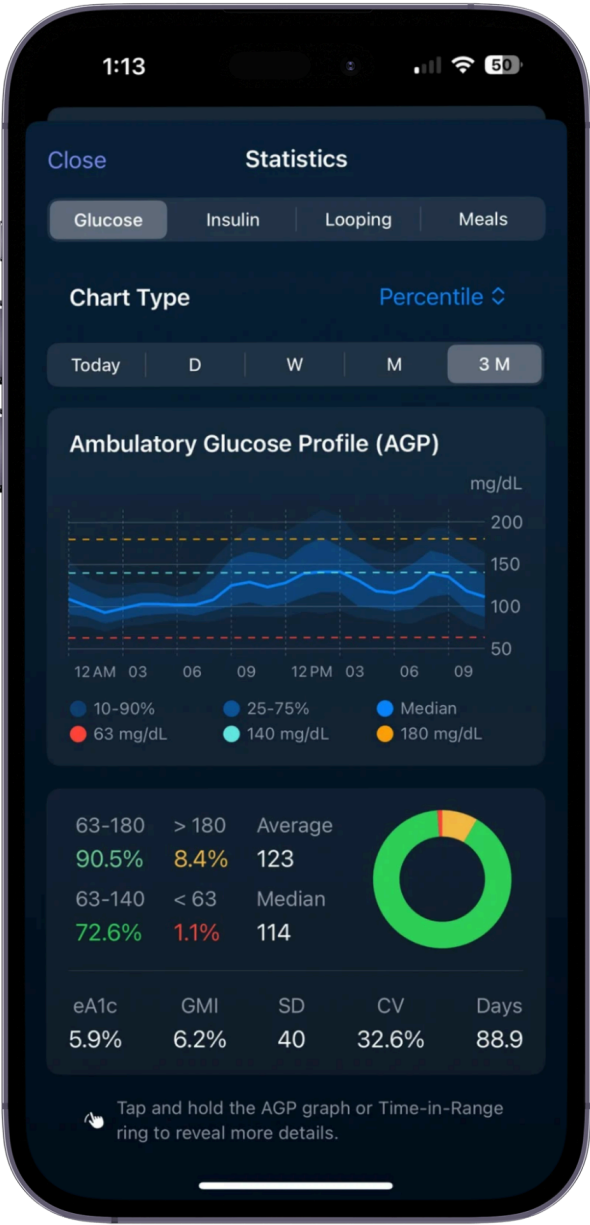
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# Statistics

# Statistics

Trio's Statistics feature provides comprehensive data analysis and visualization of your diabetes management metrics. View detailed glucose patterns, insulin usage, loop performance, and meal data across multiple time periods.



## Accessing Statistics

To view statistics:

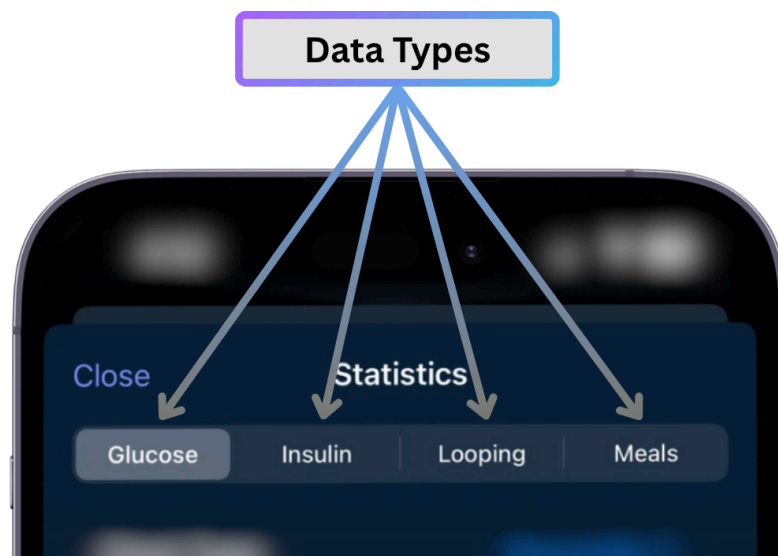
- Open Trio
- Navigate to the Statistics screen (icon in the navigation menu)
- Select a category tab (Glucose, Insulin, Looping, or Meals)
- Choose your preferred chart type and time period

Access Statistics  
Screen



## Data Types

The Statistics screen is organized into four main categories, each accessible via tabs at the top:



- **Glucose** - CGM & blood sugar metrics, time-in-range, and glucose distribution
- **Insulin** - Total daily dose and bolus distribution
- **Looping** - Loop cycle performance and reliability metrics
- **Meals** - Macronutrient tracking and meal analysis

Each category offers multiple chart types and time period options to help you understand patterns and make informed adjustments to your therapy settings.

# Glucose Statistics

## Available Metrics

The Glucose section displays seven key metrics:

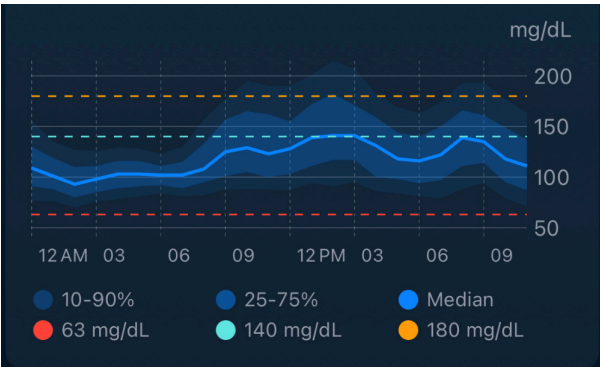
| Metric  | Description                               | Formula                                       |
|---------|-------------------------------------------|-----------------------------------------------|
| eA1c    | Estimated A1c from CGM data               | mg/dL: $\frac{Avg\ Glucose + 46.7}{28.7}$     |
| GMI     | Glucose Management Indicator              | $3.31 + (0.02392 \times Avg\ Glucose)$        |
| Average | Mean of all glucose readings              | $Sum / Count$                                 |
| Median  | 50 <sup>th</sup> percentile glucose value | Middle value when sorted                      |
| SD      | Standard Deviation (glucose variability)  | $\sqrt{(\frac{\Sigma(x - mean)^2}{(n - 1)})}$ |
| CV      | Coefficient of Variation                  | $\frac{SD}{Mean} \times 100\%$                |
| Days    | Number of days with glucose data          | Count of unique dates                         |

### Display Units

- **eA1c:** Available as % (NGSP) or mmol/mol (IFCC)
- **GMI:** Available as % or mmol/mol
- **Glucose values:** Displayed in mg/dL or mmol/L based on your settings

## Chart Types

### 1. Percentile by Time (AGP - Ambulatory Glucose Profile)



The default view shows hourly glucose percentiles across 24 hours:

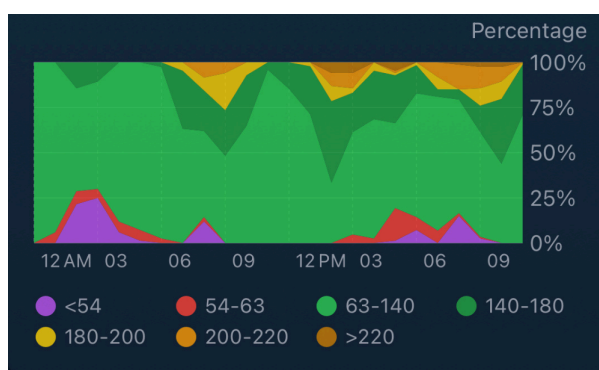
- **Dark band:** 25<sup>th</sup>-75<sup>th</sup> percentile (middle 50% of values)
- **Light band:** 10<sup>th</sup>-90<sup>th</sup> percentile (middle 80% of values)
- **Center line:** Median (50<sup>th</sup> percentile)
- **Reference lines:** Your configured low, target, and high thresholds

This chart helps identify:

- Time-of-day patterns (dawn phenomenon, overnight lows, etc.)
- Glucose variability at specific times
- How tightly controlled your glucose is

## 2. Distribution by Time

Shows the percentage of time spent in each glucose range for every hour of the day:



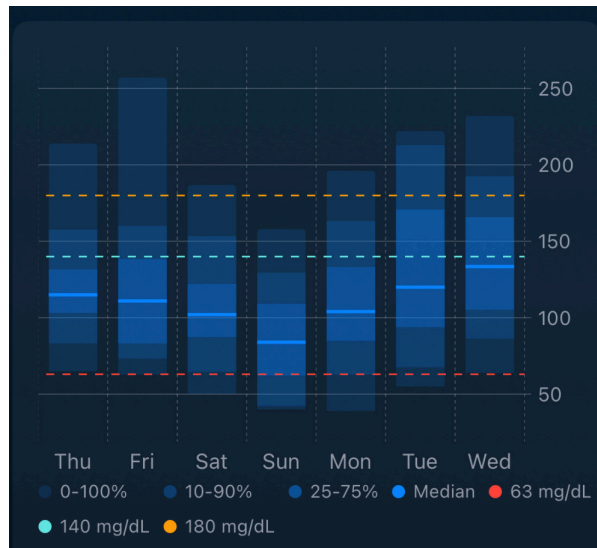
### Ranges displayed:

- **Purple:** < 54 mg/dL (< 3 mmol/L)
- **Red:** 54-70 mg/dL (3-3.9 mmol/L)
- **Bright Green:** 70-140 mg/dL (3.9-7.8 mmol/L)
- **Dark Green:** 140-180 mg/dL (7.8-10 mmol/L)
- **Yellow:** 180-200 mg/dL (10-11.1 mmol/L)
- **Orange:** 200-220 mg/dL (11.1-12.2 mmol/L)
- **Dark Orange:** > 220 mg/dL (> 12.2 mmol/L)

Use this to identify times when you're most likely to be out of range.

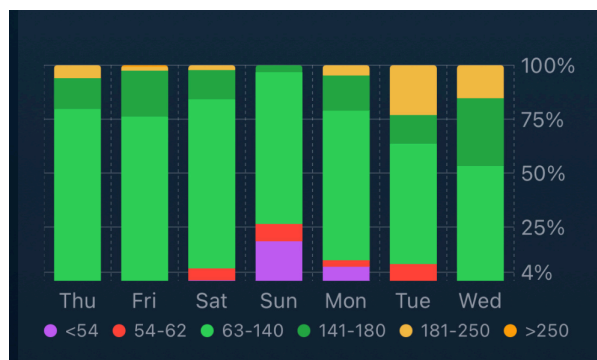
## 3. Percentile by Day

Box plot showing daily glucose distribution:



- Each day shows a box-and-whisker plot
- Box represents 25<sup>th</sup>-75<sup>th</sup> percentile
- Whiskers extend to 10<sup>th</sup> and 90<sup>th</sup> percentiles
- Center line is the median
- Compare day-to-day variability

#### 4. Distribution by Day



Stacked bar chart showing the percentage of time in each range per day:

- See which days had optimal control
- Identify patterns across multiple days
- Useful for reviewing weekly trends

Distribution by Day uses the same color chart as the [Distribution by Time](#) chart.

#### TITR vs. TING

Trio supports two tighter TIR calculation methods:

##### Time in Tight Range (TITR)

- **In Range:** 70-140 mg/dL (3.9-7.8 mmol/L)
- **Low:** 54-70 mg/dL (3-3.9 mmol/L)

- **Very Low:** < 54 mg/dL (< 3 mmol/L)

#### Time in Normoglycemia (TING)

- **In Range:** 63-140 mg/dL (3.5-7.8 mmol/L)
- **Low:** 54-63 mg/dL (3-3.5 mmol/L)
- **Very Low:** < 54 mg/dL (< 3 mmol/L)

You can select your preferred method in the Statistics settings.

#### TIR Goals

The American Diabetes Association recommends:

- **> 70%** time in range (70-180 mg/dL or 3.9-10 mmol/L)
- **< 25%** time above range (>180 mg/dL or >10 mmol/L)
- **< 4%** time below 70 mg/dL or 3.9 mmol/L
- **< 1%** time below 54 mg/dL or 3 mmol/L

## Time Periods

Select from five time period options:

- **Today:** From midnight to current time
- **Day (D):** Last 24 hours
- **Week (W):** Last 7 days
- **Month (M):** Last 30 days
- **3 Months (3M):** Last 90 days

## Insulin Statistics

### Chart Types

#### Total Daily Dose (TDD)

View your insulin usage broken down by type:

#### Components:

- **Manual Bolus:** Insulin you delivered manually for meals/corrections
- **SMB:** Super Micro Boluses delivered automatically by Trio
- **External:** Insulin injections logged but not delivered by pump
- **Temp Basal:** Insulin from temporary basal rates (above scheduled)

- **Scheduled Basal:** Insulin from your scheduled basal profile

The TDD chart displays:

- Bar graph of total daily insulin
- Statistics panel showing totals and averages for the visible range
- Scrollable timeline to review historical data

### Bolus Distribution

Detailed breakdown of bolus insulin types:

- Manual bolus totals
- SMB totals
- External insulin totals
- Average bolus amounts per period

Use this to understand:

- How much insulin comes from automatic vs. manual delivery
- SMB effectiveness and contribution to total insulin
- Bolus patterns over time

### Time Periods

Choose from four time periods:

- **Day (D):** Last 24 hours (shows hourly breakdown)
- **Week (W):** Last 7 days
- **Month (M):** Last 30 days
- **3 Months (3M):** Last 90 days

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## Looping Statistics

### Loop Performance Metrics

Track how reliably Trio is functioning:

| Metric            | Description                          |
|-------------------|--------------------------------------|
| <u>Loop Count</u> | Total number of loop cycles executed |
| Successful        | Number of successful completions     |



|                 |                                       |
|-----------------|---------------------------------------|
| Failed          | Number of failed cycles               |
| Success %       | Percentage of successful loops        |
| Median Interval | Typical time between loops (minutes)  |
| Median Duration | Typical loop execution time (seconds) |
| Glucose Count   | Number of CGM readings received       |
| Days            | Days tracked                          |

## Chart Types

### Loop Performance Chart

Bar chart showing loop execution over time:

- **Green bars:** Successful loops
- **Red bars:** Failed loops
- Hover/tap for detailed counts per time period
- Scrollable timeline

Use this to identify:

- Times when loops frequently fail
- CGM connection issues
- Overall system reliability

## Time Periods

Same as glucose statistics:

- Today, Day (D), Week (W), Month (M), 3 Months (3M)



**Good Loop Performance**

You should typically see:

- Loop interval: 5-6 minutes (Trio loops every 5 minutes normally)
- Success rate: > 95%
- Failures usually indicate CGM or pump connectivity issues

# Meal Statistics

## Macronutrient Tracking

View your nutritional intake over time:

### Macros tracked:

- **Carbohydrates:** Total grams per period
- **Fat:** Total grams per period
- **Protein:** Total grams per period

The meal chart displays:

- Bar graph showing daily or hourly totals
- Breakdown by macronutrient type
- Scrollable timeline
- Date range label

## FPU (Fat Protein Units) Support

If FPU conversion is enabled in settings, Trio tracks fat and protein to calculate their insulin impact for low-carb meals.

## Time Periods

Same as insulin statistics:

- Day (D), Week (W), Month (M), 3 Months (3M)

### Note

Meal statistics only include carbs, fat, and protein that you've logged in Trio. Accurate logging is essential for useful meal statistics.

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## Customization Options

### Time-in-Range Type:

- Select TITR (Time in Tight Range) or TING (Time in Normoglycemia)
- Changes the range definitions for TIR calculations

*Found under User Interface in Features settings*

**eA1c/GMI Display:**

- Percentage (NGSP format)
- mmol/mol (IFCC format)

*Found under User Interface in Features settings*

**Glucose Units:**

- mg/dL
- mmol/L

*Found under Units and Limits in Therapy Settings*

**Target Ranges:**

- High Limit: Default 180 mg/dL (customizable)
- Low Limit: Default 70 mg/dL (customizable)

These limits affect the color coding and range calculations in all glucose charts.

*Found under User Interface, Low and High Thresholds, in Features Settings*

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## Understanding Your Statistics

### Glucose Variability

**Standard Deviation (SD):**

- **Low SD** (< 30 mg/dL): Stable, consistent glucose
- **Moderate SD** (30-50 mg/dL): Some variability
- **High SD** (> 50 mg/dL): Significant swings

**Coefficient of Variation (CV):**

- **Target:** < 36%
- **CV < 36%** indicates well-controlled glucose variability
- **CV > 36%** suggests unpredictable glucose patterns

## Improving CV and SD

High variability often indicates:

- Inaccurate carb counting
- Incorrect ISF or carb ratios
- Inconsistent meal timing
- Exercise without appropriate adjustments

## GMI vs. eA1c

Both provide estimates of your average glucose management:

- **eA1c**: Calculated from average glucose using the ADAG formula
- **GMI**: Similar calculation, recommended by international consensus

These are **estimates** based on CGM data, not laboratory A1c measurements. They correlate well but may differ from lab results.

## TDD Trends

Monitoring your Total Daily Dose helps you understand:

- **Increasing TDD**: May indicate insulin resistance, weight gain, or illness
- **Decreasing TDD**: May indicate increased sensitivity, weight loss, or more activity
- **Stable TDD**: Consistent insulin needs

Significant TDD changes should prompt review of your basal rates, ISF, carb ratios, and Dynamic ISF Settings.

## Loop Reliability

High loop success rates (> 95%) indicate:

- Good CGM connectivity
- Reliable pump communication
- Stable Trio operation

Frequent loop failures warrant investigation:

- Check CGM sensor placement and transmitter battery
  - Verify pump is in range and functioning
  - Review phone battery and background app settings
-

## Data Availability

Statistics are calculated from data stored in Trio's database:

- **Glucose:** Requires CGM data
- **Insulin:** Requires pump event history
- **Looping:** Requires loop execution records
- **Meals:** Requires logged carb entries

If you've recently started using Trio or reset your data, some statistics may not be available until sufficient data accumulates.

### Minimum data for meaningful statistics:

- **1 day:** Today and Day views
  - **7 days:** Week view
  - **30 days:** Month view
  - **90 days:** 3 Month view
- 

## Performance Notes

Statistics calculations can be intensive for large datasets. Trio uses several optimizations:

- **Caching:** Frequently accessed calculations are cached
- **Background processing:** Heavy calculations run in background threads
- **Pagination:** Charts load data in scrollable segments
- **Concurrent calculations:** Multiple statistics computed in parallel

You may notice a brief delay when first opening Statistics or switching time periods on older devices. This is normal and ensures the UI remains responsive.

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## Troubleshooting

### "No data available" Message

If you see this message:

1. **Check data source:** Ensure you have CGM data (for glucose), logged carbs (for meals), etc.
2. **Verify time period:** Try a shorter time period if you recently started using Trio
3. **Restart Trio:** Sometimes refreshing the app resolves data loading issues

## Statistics Don't Match Other Apps

Small differences are normal due to:

- Different calculation methods (mean vs. median)
- Different time period boundaries
- Different range definitions
- Data filtering (Trio may exclude some invalid readings)

Large discrepancies should be investigated—verify your data is syncing correctly.

## Slow Performance

If statistics load slowly:

- **Reduce time period:** Use Day or Week instead of 3 Months
  - **Clear app cache:** Settings > Advanced (if available)
  - **Update Trio:** Ensure you're running the latest version
  - **Device storage:** Free up space on your iPhone
- 

## Summary

Trio's Statistics feature provides powerful insights into your diabetes management:

- **Glucose Statistics:** Understand your time-in-range, variability, and patterns
- **Insulin Statistics:** Monitor your TDD and bolus distribution
- **Loop Statistics:** Track system reliability and performance
- **Meal Statistics:** Review your macronutrient intake

Use these statistics to:

- Identify patterns and trends
- Make informed adjustments to therapy settings
- Share data with your healthcare team
- Track progress toward your diabetes management goals

Regularly reviewing your statistics helps ensure Trio is working optimally and your settings remain accurate as your insulin needs change.